Technological Forecasting & Social Change 214 (2025) 124048



Contents lists available at ScienceDirect

Technological Forecasting & Social Change

journal homepage: www.elsevier.com/locate/techfore





Comparing information completeness in brand-related user-generated content media

Sanghyun Park , Sungjoo Lee *

Department of Industrial Engineering, Seoul National University, 1 Gwanak-ro, Gwanak-gu, Seoul 08826, Republic of Korea

ARTICLE INFO

Keywords:
Brand
Media
User-generated contents
Intelligence
Comparative analysis

ABSTRACT

Due to the contemporary globalized environment, business demands have become more dynamic and complex. Companies need to capture such demands to sustain and grow further in the market, for which various sources of information have been exploited. One of these is brand-related media. UGC from the media platform includes rich information on customer opinions and feedback for products and services. Recognizing the value of UGC, previous studies have proposed an approach to extract customer needs from media platforms. However, most of these studies have focused only on a single UGC platform, whereas diverse brand-related media exist and provide heterogeneous information. To address this issue, we develop a framework to evaluate the characteristics of brand-related UGC information and compare four different UGC media, including Amazon, ProQuest, Reddit, and YouTube, in terms of how rich they are in providing semantic, sentimental, and trend information based on a quantitative approach. The research results indicate that brand-related UGC media can have their own distinguishing features in the information they provide and thus need to be carefully adopted for market research. These findings offer meaningful implications for the use of brand-related UGC to identify innovation opportunities based on market needs.

1. Introduction

With shortened product life cycles and a more globalized business environment, customer requirements for products and services have become more dynamic and complicated (Brooks, 2015). Traditional approaches to identifying customer requirements include surveys and focus-group interviews. In such approaches, only a limited number of customers are involved (Nambisan, 2002), their opinions may be biased under test conditions (Liu et al., 2019), and additional efforts are required for data collection (Pournarakis et al., 2017). However, the emergence of e-commerce and social networking sites has expanded the channels for understanding customer requirements (Bhimani et al., 2019). For example, customers express their requirements in UGC via blogs and content-sharing sites (Daugherty et al., 2008; Grosser et al., 2019; Roma and Aloini, 2019). Recently, interest in methodologies for analyzing user needs and discovering innovation opportunities through UGC has increased.

UGC consists of a large amount of unstructured data (Bazzaz Abkenar et al., 2021; Chang et al., 2019), which requires additional effort to extract valuable knowledge. In particular, a series of tasks to transform UGC data from social media into such knowledge are called social media analytics (SMA; Holsapple et al., 2018; Sivarajah et al., 2020; Zeng et al., 2010). Various SMA approaches have been proposed, which can be divided mainly into two categories: one is based on inherent structural information (e.g., structural attributes, network oriented), and the other is for analyzing content information (e.g., content oriented, opinion related, topic related; Bazzaz Abkenar et al., 2021; Stieglitz et al., 2018). Compared to traditional data sources for innovation, UGC includes rich data on customer opinions and feedback for products and services, not only of the focal firm but also of its competitors (He et al., 2016). Hence, numerous studies have proposed a framework to analyze brand-related UGC directly related to the focal firm or competitors in order to derive customer needs and identify innovation opportunities. Despite their valuable contributions, most previous studies have focused only on a single or specific media platform as a source of intelligence (Goldberg and Abrahams, 2022; Kim and Lim, 2021).

Restricting the analysis focus to a single media platform can lead to biases. For example, the topics discussed intensively may differ by platform, and topics in a broad range of fields may not be fully considered if only one media platform is analyzed. Furthermore, platforms are

* Corresponding author.
E-mail address: sungjoolee@snu.ac.kr (S. Lee).

https://doi.org/10.1016/j.techfore.2025.124048
Received 18 May 2023; Received in revised form 14 February 2025; Accepted 15 February 2025
Available online 26 February 2025
0040-1625/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

not permanent; alternative platforms must be used when data collection from one platform has become invalid or its main user group has changed. To use UGC effectively and gather useful intelligence suitable for a particular purpose, the characteristics of the information provided by each type of UGC medium must be understood clearly. We focused on the concept of "information completeness" as a data quality metric to investigate the reasons for the observed differences in analysis results across various media. Among the dimensions of information quality, information completeness indicates "the extent to which data are of sufficient breadth, depth, and scope for the task at hand" (Wang and Strong, 1996, p. 32), where "the task at hand" is to apply a variety of analytical techniques to extract meaningful information from UGC. Given that the objectives and the information required vary according to the analytical technique, the completeness of information in UGC by media appears to be assessable from multiple perspectives.

Therefore, this study aims to explore the potential bias in analysis results caused by the characteristics of information provided by various UGC media. To achieve this, we conducted a quantitative comparative analysis focusing on semantic diversity, sentiment, and trend consistency across four different platforms representing distinct media types: Amazon (e-commerce-based media), ProQuest (online mass media), Reddit (collaborative social media), and YouTube (creative-outlet social media). Through this analysis, our primary objective is to empirically demonstrate the importance of selecting appropriate media when extracting brand intelligence and to propose the necessity for companies to adopt a strategic media selection methodology that can accommodate diverse user demands.

The research results demonstrate the informational incompleteness inherent in a single media. The analysis results indicate that Amazon (e-commerce-based media) is rich in sentiment and trend consistency information. ProQuest (online mass media) generally provides positive sentiments but focuses only on a few topics. On the contrary, Reddit (collaborative social media) is characterized by negative sentiment with rich trends and semantic information. Finally, YouTube (creative-outlet social media) had the most various information, with generally positive sentiments. The remainder of this paper is organized as follows. Section 2 reviews relevant studies on brand-related UGC analytics, information completeness, and typologies of brand-related media. Section 3 explains the study's research design. Section 4 presents the analysis results based on the framework. Section 5 discusses the findings along with theoretical and managerial implications. Finally, Section 6 concludes the paper by highlighting its contribution, limitations and future research directions.

2. Background

2.1. Typologies of brand-related media

Brand-related UGC is defined by Roma and Aloini (2019) as "the type of user content created with regard to a specific brand" (p. 322). This definition comprises user reviews of brands, discussions on product (or service) functions, and comparisons among products (or services). This is expected to overcome the limitations of traditional approaches to extracting customer requirements (e.g., the high costs of data collection and limited access to customers). The research gap identified in this study involves quantitatively comparing the differences in content information characteristics among brand-related media of different types. This comparison aims to recommend the use of a combination of media or at least the selection of appropriate brand-related UGC sources that match the purpose of media content analysis. Therefore, it is necessary to first define the types of brand-related media to investigate the differences in information characteristics by brand-related media. Although no general criteria to classify brand-related media exist thus far, two representative types, social media and online communities, have been frequently mentioned in the existing literature (Goldberg and Abrahams, 2022; Xu et al., 2017).

2.1.1. Typologies of online social media

Online social media is known as a major source of UGC and is widely used for brand-related media, the characteristics of which greatly differ by type. Previous studies have proposed typologies for online social media, classifying media based on behavioral, conceptual, functional, and marketing characteristics. First, online social media was grouped with respect to their users by Brandtzaeg in 2010, based on their behavioral characteristics. The author stressed the necessity of providing common ground for explaining the various methods of using new media technologies, particularly given the more complex media environments. Consequently, eight types of media users were defined based on two criteria (i.e., variety of use and frequency of use): advanced users, debaters, instrumental users, socializers, entertainment users, lurkers, sporadics, and nonusers. Heterogeneous user requirements for each user type are expected to be considered in the development of new media.

Second, Kaplan and Haenlein (2010), who presented one of the most influential early works on the classification of online social media, categorized it based on concepts and definitions (Kaplan and Haenlein, 2010). The authors emphasized the lack of a conceptual understanding of online social media despite the increasing interest in its use in business, using two criteria—media research (i.e., social presence and media richness) and social processes (i.e., self-presentation and self-disclosure)—to offer six types of online social media: collaborative projects, blogs, content communities, social networking sites, virtual game worlds, and virtual social worlds. They also argue that the characteristics of different types must be considered when companies adopt online social media to communicate with their customers.

Third, Kietzmann et al. (2011) identified seven types of online social media functions: identity, conversations, sharing, presence, relationships, reputation, and groups. The authors insisted that managers tend to avoid or ignore online social media owing to a lack of understanding of, more specifically, what it is, which form it takes, how to participate in it, and how to learn from it. They defined the functional components of online social media and demonstrated that they could be described as various functional components. Further, the researchers analyzed user experiences from online social media and firm implications based on those functions and established firm strategies for monitoring and responding to diverse online social media activities.

Finally, Zhu and Chen (2015) classified online social media based on two marketing-related criteria: nature of the connections (i.e., profile-based vs. content-based) and customization level of messages (i.e., broadcast vs. customized), thereby producing four types of online social media: relationships, self-media, collaboration, and creative outlets. The authors argue that online social media marketing is only effective when marketing efforts correspond to and are adjusted to the diverse requirements of online social media users. They also suggest that a massive investment in online social media advertising does not always positively affect a consumer's decision; accordingly, they provide valuable implications for online social media marketing. We referenced the work by Zhu and Chen (2015) in determining the types of media-related UGC targeted for further analysis because brand-related UGC is closely related to marketing-related aspects. More specifically, profile-based social media concerns information about users, whereas content-based social media is more for content.

2.1.2. Brand-related media outside of online social media

Brand-related UGC also exists outside of social media. In particular, because relying solely on UGC from online social media can provide limited insights (Mariani and Nambisan, 2021), it is necessary to pay attention to a more diverse range of media types. Traditionally, user review data present on online review platforms is a major source of information for brand-related media. *E*-commerce-based media refers to media integrated within e-commerce platforms. Product reviews are a typical form of UGC for e-commerce-based media. Because users' communication and information-sharing functions are limited, e-

commerce-based media are not considered online social media. Amazon and Booking.com are examples of e-commerce-based media, where various mechanisms exist to assess the quality and reliability of content. Through the analysis of data available on these platforms, businesses can gain valuable insights to aid in strategic decision-making (Mariani and Nambisan, 2021).

Expanding the concept of brand users beyond traditional buyers or users, to include experts, influencers, journalists, and so on, allows for the extraction of diverse brand insights through a broader range of media. From this perspective, online mass media (e.g., New York Times or Washington Post) can be another valuable source of brand-related UGC. Online mass media emerged with the development of search engines that enabled the retrieval of articles published by numerous presses or magazine publishers, making it possible to find articles of interest from the vast amount of mass media articles nowadays. Therefore, we consider two more types of nononline social media platforms worth extracting brand-related intelligence, including e-commerce-based media and online mass media.

2.2. Brand-related media analytics

Brand-related media analytics indicate a series of activities required to transform brand-related UGC into valuable knowledge to support business processes. First, we review the approaches for extracting intelligence from UGC data to understand how UGC data have been analyzed for different purposes. Because the brand-related UGC is in the form of a large volume of unstructured text data, natural language-processing techniques and various other analysis techniques are applied to convert the data into intelligence. We focused on studies that developed an analysis framework and applied it to brand-related UGC. In these studies, various brand-related UGC platforms have been adopted using different approaches, as summarized in Table 1. According to Stieglitz and Dang-Xuan (2013), social media analytics can have three approaches: topic-related, opinion-related, and structural attribute approaches. The topic-related approach examines the issues discussed regarding a target product or service by applying techniques such as content analysis or text mining to UGC. Using sentiment analysis, an opinion-related approach monitors customer satisfaction and reactions to a target product or service. The structural attribute approach analyzes user relationships and identifies user communities through a social network analysis.

Table 1
Brand-related media used as a case study in previous research.

Publication	Media	Approach (methods)		
		Topic modeling	Opinion mining	Time-series analysis
Pournarakis et al., 2017	Twitter	LDA(Latent dirichlet allocation)	SVM(Support vector machine)	Used
Tsirakis et al., 2017		–	SVM	Used
Amirmokhtar Radi and Shokouhyar, 2021		Heuristic algorithm	Lexicon-based	Used
Reyes-Menendez et al., 2020		LDA	SVM	–
Zhan et al., 2021		LDA	Lexicon-based	–
Mostafa, 2013			Lexicon-based	Used
Jeong et al., 2019	Reddit	LDA	Deep learning-based	
Choi et al., 2020		Event detection and tracking algorithm	Lexicon-based	Used
Wang et al., 2018	Amazon	LDA	–	–
Goldberg and Abrahams, 2022		Heuristic algorithm	–	–
Lee et al., 2022	Trip-adviser	–	Lexicon-based	–
Chang et al., 2019		Convolution tree kernel	Convolution tree kernel	Used
Xu et al., 2017	Booking	Latent semantic analysis	Regression analysis	–
Jelodar et al., 2021	YouTube	LDA	Fuzzy lattice reasoning	–

Here, the structural attribute approach was not considered because of its limited availability in providing brand-related information and presence of information varies across different media. Instead, we added another type of approach—the trend-related approach—given that most brand-related UGC is time-series data. Consequently, the existing studies were reviewed in terms of the approaches taken to analyze the brand-related UGC data, which show that both online social media platforms (e.g., Twitter, Facebook, Reddit) and online review platforms (e.g., Amazon, TripAdvisor) have mainly adopted the three approaches for brand-related media analytics.

2.3. Comparative analysis of information differences across media

Customers who use brand-related media cannot represent all customers. The use of data lacking representativeness, because online social media does not reach all customers, can put companies at risk (Lee, 2018). Moreover, considering the impact of the evolving social media environment on brand-related UGC, it is necessary to consider multiple sources for an overall product evaluation (Roma and Aloini, 2019).

In light of these issues, there are studies aimed at quantitatively understanding the differences in informational characteristics of content across various media. Jang and Yoon (2018) quantitatively analyzed the differences in content information between online news and social network sites. Through simple keyword analysis, they found that news content is event focused, continuous, and specific, whereas content on social network sites is centered around personal interests and the topics change rapidly (Jang and Yoon, 2018). This study is significant for its comparative analysis of the informational differences between two types of media from the perspectives of content and trends. However, it has the limitation of relying on keyword analysis that fails to account for context in its quantitative comparison.

In the study by Zhang et al. (2019), opinion mining was applied to data from more than two sources of book reviews to identify that content on e-commerce sites mainly discusses external aspects of books like packaging, whereas social network sites tend to focus on the story of the books. The research suggests that due to these differences, evaluating products should consider multiple sources rather than relying on data from a single source for a comprehensive assessment (Zhang et al., 2019). This study is valuable for designing various metrics to compare and analyze the informational differences between media in terms of content and sentiment. However, it has the limitation of focusing solely on results in its analysis, lacking theoretical basis for the causes of observed differences.

Therefore, this study aimed to perform a comprehensive comparison of topic, time series, and sentiment aspects, as previously intended by other researchers. It focused on the concept of "information completeness" as a data quality metric to investigate the reasons for the observed differences in analysis results across various media. Furthermore, for a more objective comparison, this study utilized a transformer-based pretrained model to quantify each informational characteristic before conducting a comparative analysis.

3. Research design

3.1. Analysis design

An analysis design was developed to empirically evaluate the differences in information completeness among brand-related media, as shown in Fig. 1. We first collected data from multiple brand-related

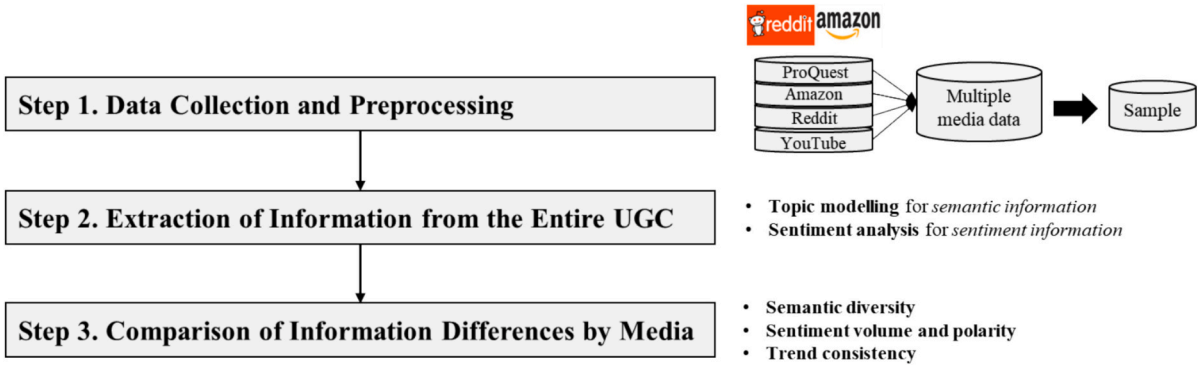


Fig. 1. An analysis design.

media. The information completeness was then analyzed from three perspectives – semantic diversity, sentiment abundance, and trend consistency – for which BERTopic, sentiment analysis, and autocorrelation index function (ACF) analysis were conducted. The analysis results were then compared to evaluate the degree of information completeness, based on which strategic and managerial implications for using brand-related media were provided.

To verify the significant differences in the information that can be extracted from UGC across media types, we selected various types of media for analysis. Consequently, we selected media with sufficient brand-related UGC to allow for meaningful analysis. Given that this study's focus is brand-related UGC, we expect that more relevant information is provided in content-based than in profile-based social media. Content-based UGC is classified into two categories, depending on the customization level of messages: collaborative social media and creative outlet social media (Zhu and Chen, 2015). As a result, Reddit (collaborative social media) and YouTube (creative outlet social media) were selected. Although online social media is a significant source of UGC, it is important to note that not all UGC in online social media pertains to brands. Amazon is recognized as the largest e-commerce site, in which reviews from previous purchasers are made available for potential customers to reference before making a purchase (Haque et al., 2018). Meanwhile, ProQuest launched a major newspaper digitization project in 2001, creating a comprehensive academic and publication database that includes a variety of renowned publishers, such as The New York Times, The Wall Street Journal, and The Washington Post (James-Gilboe, 2005). Consequently, we selected Amazon (an e-commerce-based medium) and ProQuest (an online mass media platform) as subjects for comparison. Additionally, to ensure a fair comparison, we controlled for variables such as period, brand, and product category.

3.2. Selection of comparative dimensions and hypothesis setting

The criteria for selecting information dimensions exist irrespective of the type of brand-related media for comparison and can be measured solely by content and date of content creation, which are widely utilized in UGC methodologies. We explored three approaches to extracting information from UGC using social media analytics: topic modeling, opinion mining, and trend analysis. For each approach, we identified the semantic diversity factor, sentiment abundance, and trend consistency as the information dimensions that significantly influence the analysis results. Fig. 2 shows the three information dimensions derived using these three approaches. It should also be noted that the analysis was limited to text. Although UGC content can take various forms, such as pictures and videos, texts are the richest in content.

First, the semantic diversity factor stresses the information extracted following the topic-related approach. The semantic diversity factor refers to topics created from various perspectives of brand perception. The accuracy of information is as important as the value of information guaranteed by diversity (Ogunseye and Parsons, 2018), and this can facilitate new discoveries while leading to unexpected insights (Parsons and Wand, 2014). Completeness of semantic diversity factor in terms of information completeness means that it fully reflects the user's multifaceted perspective on the brand. If the semantic diversity factor's completeness is sufficient, companies can understand users' needs from various perspectives, leading to various innovative ideas. Conversely, from the perspective of semantic diversity, if the information is incomplete, interpretation from multifaceted viewpoints may not be possible. We have posited the following hypothesis, assuming that there is a difference in the level of semantic diversity across media and that the themes of information that primarily appear will differ.

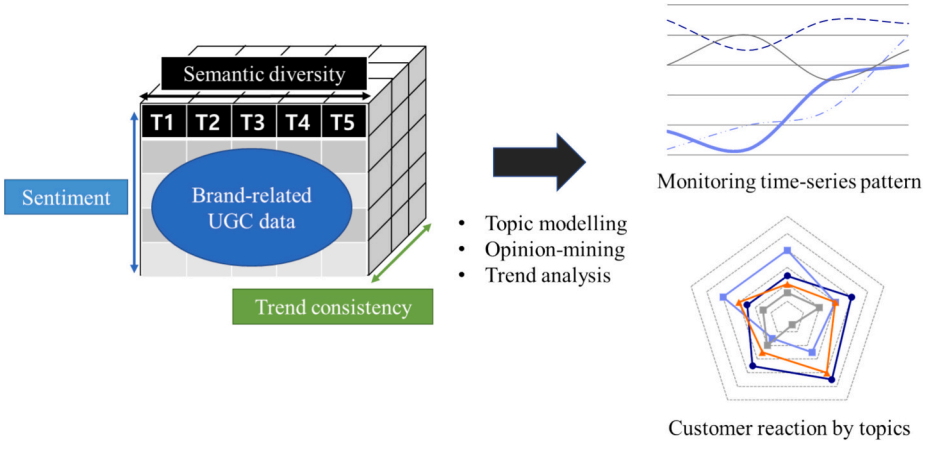


Fig. 2. Approaches for independent information extraction.

Hypothesis 1. There are differences in the information each medium presents regarding semantic diversity.

Second, the sentiment factor concerns information that can be derived following the opinion-related approach. The sentiment factor responds to a user's reaction to a brand, which allows a company to assess its brand. Sentiment analysis is better suited for subjective opinions than objective facts (Ko et al., 2017); therefore, the detection of document subjectivity is considered a necessary step before sentiment analysis (Chaturvedi et al., 2018). The sentiment factor's completeness in terms of information completeness implies that there is sufficient reaction information for various users. If the sentiment factor's completeness is insufficient, it can lead to inaccurate judgments regarding user needs. We assume that because each medium possesses a different amount of sentimental information, some media will be overall better suited for sentiment analysis. Moreover, we have posited the following hypothesis, assuming that even for the same topic, the amount of sentimental information each medium contains differs and could elicit different responses.

Hypothesis 2A. In the dimension of sentiment, there exists a difference in the volume of overall sentiment information across media.

Hypothesis 2B. In the dimension of sentiment, there exists a difference in the volume of topic-specific sentiment information across media.

Hypothesis 2C. In the dimension of sentiment, there exists a difference in the polarity of topic-specific sentiment information across media.

Third, the trend consistency factor is associated with the analysis of trend-related approach. The trend consistency factor refers to the predictable time-series change pattern information for a brand, which allows a company to monitor changes in user needs for its brand. If a consistent trend exists between two adjacent time points, there is a higher likelihood of detecting trends that do not actually exist (Hamed and Rao, 1998). In terms of information completeness, the completeness of the trend consistency factor implies the existence of trends in the frequency of UGC by topic and consumer responses. We assume that, even for the same topic, there will be media in which content consistently emerges and media in which content is generated due to specific events. Furthermore, we assume that there will be media in which consumer responses consistently appear, making them suitable for monitoring changes.

Hypothesis 3. There are differences in trend consistency by topic across media.

3.3. Methodology

3.3.1. Data collection and preprocessing

First, we conduct the data collection and preprocessing stages in order to perform a comparative analysis. It is necessary to control for conditions other than media (e.g., time period, brand, product line) for comparison. For the time period condition, it is important to standardize the timeframe across the media being compared. In terms of the brand condition, since the input required for data collection varies by media, efforts are made to standardize this input as much as possible. For example, social media platforms like Reddit can provide data related to specific brands through dedicated subchannels, while e-commerce platforms like Amazon can gather reviews for individual products. Therefore, it is essential to carefully select the scope of brands and product lines for analysis.

The collected data may differ in form depending on the media, so in cases of video or audio, it is converted into text. Depending on the media, individual documents may consist of a single paragraph, multiple paragraphs, or extended scripts. To normalize the information content in each document, all individual documents are segmented into sentence-level units. Because UGC data can contain a significant amount of noise, a step to remove this noise is undertaken. Finally, for quantitative comparison, the data are categorized according to controlled characteristics (e.g., period, product line), and the stratified sampling of the same size is applied to construct the comparison dataset.

3.3.2. Extraction of information from the entire UGC

Initially, the BERTopic method is applied to extract semantic information from UGC. BERTopic is a topic modeling technique that combines sentence vectorization, dimensionality reduction, and clustering methods to generate topics while preserving semantic information. Notably, as the unit of embedding is the sentence and both dimensionality reduction and clustering techniques can be substituted as needed, it allows for the application of state-of-the-art methods according to specific requirements (Grootendorst, 2022).

Subsequently, to extract sentiment information from UGC, sentiment analysis techniques are applied. For this purpose, models capable of distinguishing between negative, positive, and neutral sentiments for measuring sentiment density are chosen. For sentiment analysis, rule-, machine learning–, and neural network–based approaches can be applied. Considering the classification performance and the fact that the data are from media sources, it is appropriate to apply a neural network–based sentiment analysis model that has been pretrained on media data. These models are trained by adjusting document parameters through the application of backpropagation and optimization algorithms, grounded on a vast amount of labeled embedded text. Sentiment analysis models are generally trained on large volumes of social media data or review data, making them suitably applicable to this experiment, which utilizes similar data types.

3.3.3. Comparison of information differences by media

To quantitatively compare the semantic differences of UGC across various media, this study employs the probability distributions of topics derived from each source. Subsequently, to evaluate the completeness on the "semantic diversity" dimension, the diversity of topic distributions is measured. From the perspective of information completeness, the ideal media for topic modeling are those with high diversity. A narrow focus on a limited number of topics may restrict insights, whereas a diverse range of topics enables capturing demand across different areas. Therefore, diversity is assessed using the Shannon diversity index, which takes into account species richness and evenness (Shannon, 1948) (see Eq. 1).

$$\textit{Semantic diversity} = -\sum_{i=1}^{s} p_i \ln(p_i) \tag{1}$$

where p_i = *Ratio of specific topic i to all topics*, s = *Total number of topics*

To compare the differences in sentiment information of UGC across various media, this study analyzes two sub-indicators: the quantity of sentiment information and its extremity. The quantity of sentiment information is measured by the sentiment volume, which assesses the ratio of sentiment information within the content. This measure ranges from 0 to 1, with values approaching 1 indicating a higher proportion of sentiment information. However, even if the quantity of sentiment information is similar across different media, differences in consumer reactions can lead to incorrect conclusions if analysis is based solely on a single medium. Therefore, the extremity of sentiment is also compared. The sentiment polarity indicates the direction of sentiment information, ranging from −1 to 1. Values close to 1 represent positive sentiment, whereas values near −1 indicate negative sentiment (see Eq. 2).

$$0 \le \textit{Sentiment volume}_k = \frac{V_{k_NEG} + V_{k_POS}}{V_{k_NEG} + V_{k_POS} + V_{k_NEU}} \le 1 \tag{2}$$

$$-1 \le \textit{Sentiment polarity}_k = \frac{V_{k_POS} - V_{k_NEG}}{V_{k_NEG} + V_{k_POS}} \le 1$$

where $V_{k_{NEG}}, V_{k_{POS}}, V_{k_{NEU}}$ = *Document size of topic k classified as Negative*,

Positive, Neutral

Finally, the trend consistency factor was extracted by applying the autocorrelation function (ACF) to assess the stationarity within the time-series data of UGC. The ACF is commonly utilized to evaluate the stationarity of time-series UGC data, as it represents the correlation between time t and time t-k. This method allows for the quantitative evaluation of trend information in the data (see Eq. 3). The ACF can be used to assess whether the value at a specific point in time is strongly correlated with the value at a previous time point. A high level of autocorrelation suggests a consistent pattern or trend in the data. The ACF ranges from −1 to 1, with positive values indicating a positive correlation (the current value is proportional to the previous value) and negative values indicating a negative correlation (the current value is inversely proportional to the previous value).

$$ACF(k) = \frac{\sum_{t=k+1}^{T} \left(y_t^i - \overline{y^i}\right)\left(y_{t-k}^i - \overline{y^i}\right)}{\sum_{t=1}^{T} \left(y_t^i - \overline{y^i}\right)^2} \quad (3)$$

where y_t^i = *Volume of topic i at time t*; y_{t-k}^i = *Volume of topic i at time t − k*;

$\overline{y^i}$ = *The average of the volume of topic i*; T = *The end of time series*

4. Results

4.1. Data collection and preprocessing

To enable a quantitative comparison among different media information, controlling the brand and product lines within brand-generated content (UGC) is necessary. We aimed to select product lines that are sufficiently technologically intensive to enable users to generate a variety of content, have shown relatively recent growth with significant potential for improvement, and are of a scale that allows for comparative analysis. Consequently, this study has chosen tablet PCs as the subject of the case study. A tablet is a device based on a touchscreen display (sometimes along with a virtual keyword), which enables users to enter data easily. It generally has a diagonal size greater than five inches and is linked to the Internet via wireless network services. Although it shares much in common with smartphones, a tablet is differentiated from a smartphone because its calling function is not prioritized. According to a 'Tablet vendor market share worldwide' by Statcounter (2023),[1] the top brand in the tablet market is Apple (53 % market share), followed by Samsung (28 % market share). Due to Apple's market share exceeding 50 %, making it the highest among competitors, brand-related media data from Apple has been adopted as the case study subject for this research.

We collected English-language UGC data on Apple tablet PCs from January 2019 to December 2023, spanning different characteristics of brand-related media. All posts written in languages other than English and those with fewer than five words were excluded. Initially, UGC data from collaboration-type online social media were collected from Reddit, specifically from subreddits related to Apple "iPad," yielding 30,662 documents. Secondly, creative outlet-type online social media UGC data were obtained through the metadata and subscription data of YouTube videos, with a yearly search using the query "Apple iPad tablet" on the YouTube AP, resulting in 880 UGC documents. Thirdly, e-commerce-based media UGC data were collected from 'Amazon.com' by searching for "Apple iPad," selecting related products (152 in total), and gathering up to 50 reviews per product, resulting in a total of 4374 documents. Finally, online mass media UGC data were compiled from ProQuest articles, with 1819 articles featuring "Apple" and "iPad" in their titles, collected from a total of 157 publishers. The initial number of documents used for analysis is attached below as Fig. 3.

Due to the varying amount of information per document across different media, sentence segmentation was performed. The Python library pySBD[2] was utilized for sentence segmentation (Sadvilkar and Neumann, 2020). Following segmentation into sentences and filtering for sentences with more than five words, the results were as follows: Amazon had 13,990 sentences, ProQuest had 46,878 sentences, Reddit had 252,854 sentences, and YouTube had 61,877 sentences. Stratified sampling was then conducted for brand-related media based on the period and the minimum number of sentences corresponding to brands. This approach aimed to balance the ratio of all brands, ultimately producing a set of 42,420 sentence documents. The comparative data and stratified sample results are organized in Fig. 4 below.

4.2. Extraction of information from the entire UGC

The Python BERTopic[3] package was adopted to categorize the sampled 42,420 documents into semantic information. The open-source 'General Text Embeddings model,' vectorizing all sentences into 1024 dimensions, was the sentence-transformer model used. This dimensionality was chosen for the sentence-transformer model to facilitate subsequent clustering tasks by utilizing the relatively low dimensionality of 1024 dimensions from the 'General Text Embeddings model' (Li et al., 2023), thereby vectorizing all sentences. Dimensionality reduction and clustering tasks were then carried out using UMAP and HDBSCAN, the default models of BERTopic. In terms of hyperparameter tuning, it was found that the results were significantly affected by the min_cluster_size parameter of HDBSCAN. Therefore, after evaluating the average DBCV score and Cluster_persistence values, the min_cluster_size was set to 340, and the min_sample was set to 1. A total of 28 topics were generated, including one outlier topic. Four topics that appeared in >90 % of a single media type were excluded. Some of the results are presented in Table 2 below.

The sentiment information factor was extracted using a transformer-based pretrained sentiment labeling model. The Twitter-Roberta-base for Sentiment Analysis was selected as the model for analysis, a sentiment model trained to classify texts into three categories – negative, neutral, and positive–using 58 million tweets (Loureiro et al., 2022). The sentiment analysis results are summarized in Appendix A. After classification, the sentiment volume and polarity for each topic were calculated.

4.3. Comparison of information differences by media

The analysis results were visualized using a heatmap to assess the characteristics of the information. BERTopic modeling was applied to four types of UGC data, with each document being assigned to one of 23 topics. A chi-square test conducted on the frequency of occurrence of these 23 topics across the four media types showed that, at a significance level of 0.05, the *p*-value was 0.000, and the chi-square statistic was 15,320.742, indicating a significant difference across groups. Following the application of the Bonferroni method, it was found that at an adjusted significance level of 0.008, a difference exists in the frequency of topics appearing across all media types. ProQuest and Reddit were the two media types with the most different distributions of topics, while Reddit and YouTube seemed to have the most similar topic distributions. Detailed results of the post-hoc analysis are attached below in Table 3.

To identify differences in the main topics emerging across various media, we visualized the proportion of each media type per topic using a heatmap. Fig. 5 illustrates a heatmap for the semantic information

[1] Tablet Vendor Market Share Worldwide, https://gs.statcounter.com/vendor-market-share/tablet/worldwide/

[2] pySBD Library, v0.3.4, https://pypi.org/project/pysbd/

[3] Python BERTopic library, v0.16.0, https://pypi.org/project/bertopic/

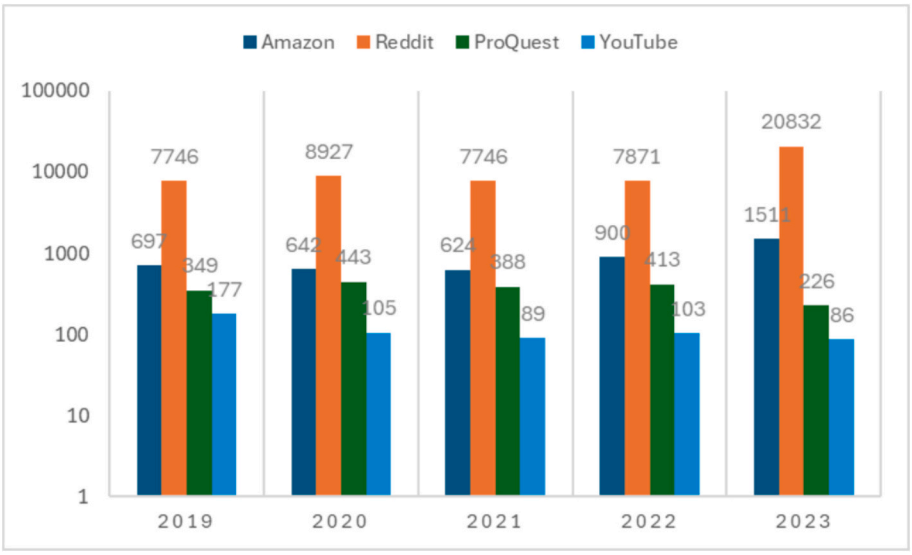


Fig. 3. Number of documents collected by media from 2019 to 2023.

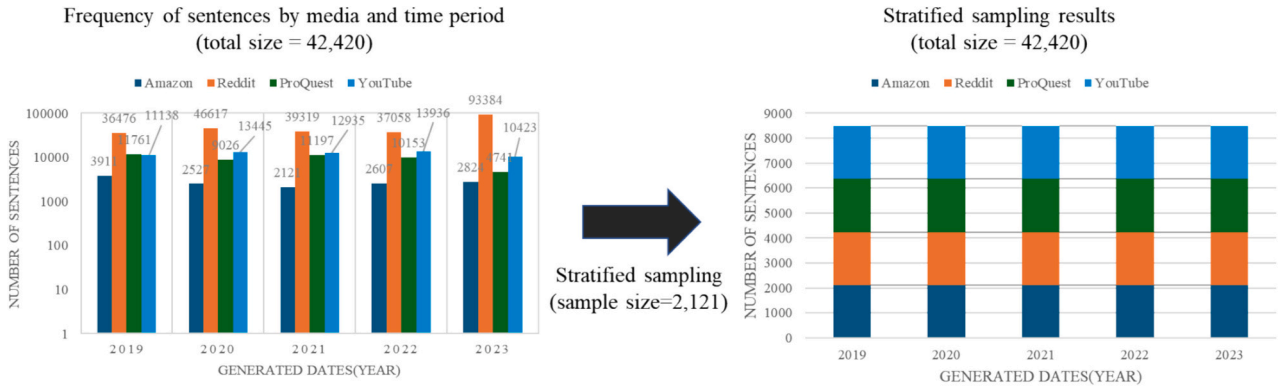


Fig. 4. Results after applying stratified sampling following sentence segmentation.

factor. This approach allowed us to see how different media prioritize specific topics, indicating each platform's unique informational focus or user interest. The result revealed significant differences in the range of topics UGC covers depending on the platform. For instance, Amazon contained the most information on Topic 1 (Battery and Charging), Topic 16 (Customer Service and Returns Process), and Topic 22 (Tablet Gifts for Family Celebrations). Conversely, Reddit was found to have the most information on Topic 8 (Competitor Characteristics) and Topic 23 (Technical troubleshooting). Consequently, Hypothesis 1 (There are differences in the information each medium presents regarding semantic diversity) was supported.

Next, the proposed Eq. 1 was calculated to compare the semantic diversity across different media. While Reddit exhibited the highest semantic diversity for the top 5 topics based on frequency, YouTube showed greater semantic diversity starting from the top 10. Detailed results of the semantic diversity results are attached below in Table 4.

Secondly, upon conducting a chi-square test for the sentiment analysis results across the four types of media, it was found that at a significance level below 0.05, the *p*-value was 0.000, and the chi-square statistic was 2792.447, indicating a significant difference in sentiment frequencies across the groups. Following the application of the Bonferroni correction, at an adjusted significance level below 0.008, a difference in the frequency of sentiments appearing across all media was observed. Consequently, Hypothesis 2A (In the dimension of sentiment, there exists a difference in the volume of overall sentiment information across different media) was supported.

Next, to compare differences in sentiment volume by topic, a Kruskal-Wallis test was performed using Eq. 2 to evaluate the differences in sentiment volume by topic. The results showed a *p*-value of 0.000 and a chi-square statistic of 50.244, indicating a significant difference in the average sentiment volume by topic. Subsequent individual comparisons conducted after applying the Bonferroni correction revealed that only Amazon showed a significant difference in the average sentiment volume by topic. Detailed results of the semantic volume results are attached below in Table 5.

Figs. 6 and 7 show the evaluation results utilizing heatmaps for the sentiment information factor. The value in Fig. 6 represents the volume of sentiment information (D_k). Amazon exhibited the highest volume of sentiment information, with D_k values ranging from 0.43 to 0.73, while the other three media types displayed a similar level of sentiment information volume. The observed difference in the average volume of sentiment by topic, with Amazon generally having a higher sentiment volume, can be attributed to its overall higher sentiment volume. However, the heatmap below suggests apparent differences in sentiment volume among the remaining three media types. Consequently, Hypothesis 2B (In the dimension of sentiment, there exists a difference in the volume of topic-specific sentiment information across different media) was only partially supported.

Following the calculation of the differences in sentiment volume by topic using Eq. 2, a Kruskal-Wallis test was conducted. The results

Table 2
Results of topics classified using BERTopic.

Topic	Count	Label	Representation
0	5846	Overall displays	['refurbished', 'bionic', 'oled', 'retina', 'imac', 'bezels', 'chips', 'upcoming', 'displays', 'gpu']
1	2167	Battery and charging	['charger', 'plugged', 'chargers', 'watt', 'adapter', 'dies', 'brick', 'aftermarket', 'batteries', 'cords']
2	1542	Advanced tablet input accessories	['trackpad', 'folio', 'logitech', 'typing', 'keyboards', 'magnets', 'kickstand', 'combo', 'gestures', 'touchpad']
3	1466	OS security updates	['beta', 'macos', 'siri', 'vulnerabilities', 'bug', 'fixes', 'developers', 'watchos', 'webkit', 'hackers']
4	1452	Simple troubleshooting	['reset', 'swipe', 'multitasking', 'widgets', 'window', 'buttons', 'restart', 'splitscreen', 'reboot', 'freezes']
5	1443	Digital art and note-taking tools	['drawing', 'paper', 'onenote', 'notetaking', 'handwriting', 'notebook', 'wacom', 'folders', 'handwritten', 'sketching']
6	1330	Entertainment and gaming	['xbox', 'controller', 'netflix', 'student', 'browsing', 'consumption', 'pubg', 'kindle', 'playstation', 'console']
7	1290	Tablet market trends and sales	['billion', 'sales', 'september', 'manufacturing', 'china', 'wwdc', 'smartphone', 'demand', 'cupertino', 'airpods']
8	1279	Competitor brands	['lite', 'dex', 'pen', 'samsungs', 'spen', 'lte', 'amoled', 'tradein', 'sfe', 'yellow']
10	1156	Digital drawing and sketching tools	['pen', 'stylus', 'nibs', 'pencils', 'draw', 'drawing', 'wacom', 'accessory', 'crayon', 'sketching']
11	945	Feedback and advice	['suggestions', 'reddit', 'opinions', 'feedback', 'leo', 'english', 'posting', 'recommendation', 'installation', 'petty']
12	880	Advanced tablet camera	['megapixel', 'cameras', 'lidar', 'sensor', 'landscape', 'ultrawide', 'selfie', 'frontfacing', 'augmented', 'photography']
13	777	Seasonal shopping and deals	['deals', 'discounts', 'shopping', 'retailers', 'currys', 'walmart', 'thanksgiving', 'lewis', 'discount', 'appliances']
16	642	Customer service and returns process	['seller', 'refund', 'customer', 'contacted', 'returning', 'refunded', 'returned', 'hassle', 'shipped', 'returns']
17	611	Displays (in detail)	['hz', 'brightness', 'hertz', 'displays', 'bezels', 'oled', 'lcd', 'pixels', 'amoled', 'colour']
18	610	Compact and portable technology	['factor', 'report', 'minis', 'compact', 'jelly', 'macrumors', 'portability', 'smallest', 'consumers', 'specifications']
19	563	Connectivity solutions and accessories	['usbc', 'bluetooth', 'hub', 'dongle', 'cables', 'headset', 'dock', 'adapters', 'connectors', 'charges']
20	423	Cellular network options	['sim', 'cellular', 'hotspot', 'verizon', 'connectivity', 'tmobile', 'att', 'unlocked', 'carriers', 'tethering']
21	421	Storage options	['gigabytes', 'gigabyte', 'cellular', 'capacity', 'terabyte', 'microsd', 'pesos', 'micro', 'ssd', 'expandable']
22	404	Tablet gifts for family celebrations	['christmas', 'birthday', 'grandma', 'enjoying', 'grandkids', 'daughters', 'olds', 'wants', 'autism', 'xmas']
23	360	Technical troubleshooting	['workaround', 'solve', 'bug', 'encountered', 'workarounds', 'troubleshooting', 'fixes', 'defective', 'errors', 'glitch']
24	354	Backup and recovery strategies	['icloud', 'itunes', 'backup', 'locked', 'recovery', 'passwords', 'appleid', 'unable', 'syncing', 'activation']
25	351	Creative software	['cut', 'logic', 'lumafusion', 'lightroom', 'photoshop', 'footage', 'imovie', 'workflow', 'recording', 'keyframing']

Table 3
Results of pairwise comparisons with the frequency by topic as the independent variable.

Compare	Chi-square statistic	Degrees of freedom	Adjusted alpha	Original P-value
Amazon-ProQuest	4619.829	22	0.008	0.000(***)
Amazon-Reddit	4371.627	22	0.008	0.000(***)
Amazon-YouTube	3038.875	22	0.008	0.000(***)
ProQuest-Reddit	5596.487	22	0.008	0.000(***)
ProQuest-YouTube	2843.884	22	0.008	0.000(***)
Reddit-YouTube	1812.807	22	0.008	0.000(***)

showed a *p*-value of 0.000 (***), and the chi-square statistic was 21.716, indicating a significant difference in the average sentiment polarity by topic. Subsequent individual comparisons, conducted after applying the Bonferroni correction, revealed that a significant difference in the average sentiment volume by topic was only observed between 'ProQuest-Reddit' and 'Reddit-YouTube.' Detailed results of the semantic polarity results are attached below in Table 6.

The values in Fig. 7 represent the polarity of the sentiment information (P_k). We observed a significant difference in polarity according to platform and topic. For example, Proquest and YouTube are almost positive in their descriptions of the target brand, regardless of the topic. In contrast, Amazon is various, and Reddit is virtually negative. It seems that the positive aspects of the brand are emphasized on YouTube and ProQuest, while Reddit is rather negative. Given the high density of sentiment information, Amazon is neutral, in which both negative and positive aspects are discussed. Consequently, Hypothesis 2C (In the dimension of sentiment, there exists a difference in the polarity of topic-specific sentiment information across different media) was supported.

Finally, to compare differences in trend consistency across media, quarterly trend consistency was calculated for the top 11 topics with total occurrences exceeding 1000, using Eq. 3. Conducting a Kruskal-Wallis test with topic-specific trend consistency as the independent variable resulted in a p-value of 0.016 (**), and the chi-square statistic was 10.283, indicating a significant difference in the average trend consistency by topic at a significance level below 0.05. Subsequent individual comparisons, conducted after applying the Bonferroni correction, showed that none of the pairwise comparisons were significant at the adjusted significance level below 0.008.

A heatmap was constructed to compare differences in topic-specific trend consistency, as shown in Fig. 8. The values in the figure represent the AutoCorrelation Function (ACF), indicating the degree of trend information. A clear distinction between platforms and topics was observed. It was noted that Amazon and Reddit, in particular, consistently featured some topics, indicating relatively high trend consistency. Consequently, Hypothesis 3 (There are differences in trend consistency by topic across media) was supported.

5. Discussion

5.1. Summary of findings

The results of this analysis are summarized in Table 7. Amazon (a platform for e-commerce-based media) is rich in sentiment and trend information but has narrow semantic information. First, given the nature of e-commerce-based media, UGC will likely have positive and negative opinions from this platform about the products and services consumers have bought. Consequently, the content is limited to several key topics within a period when products or services are sold. Thus, this platformis useful for identifying understand the market needs of target

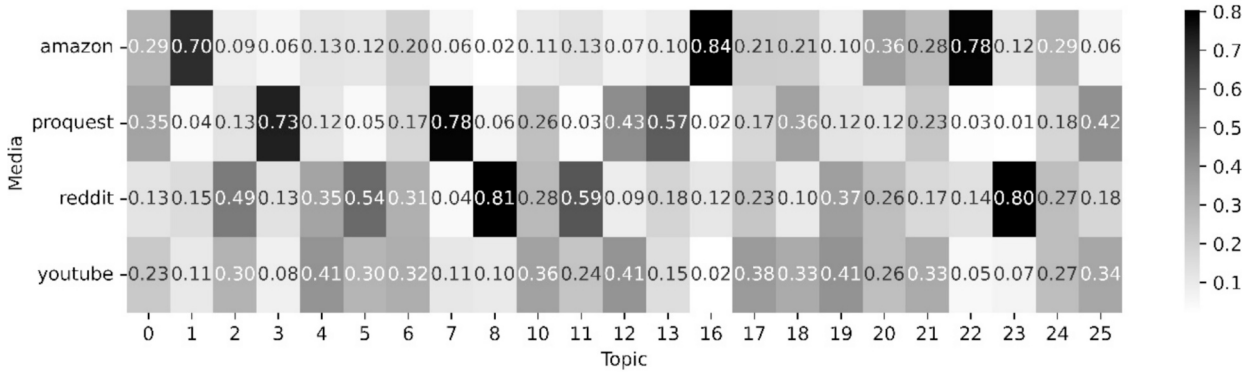


Fig. 5. Semantic diversity information completeness heatmap by brand-related media.

Table 4
Results of semantic diversity across media by the top N of topics.

Media	Semantic diversity for top N topics by frequency				
	Top 5 topics	Top 10 topics	Top 15 topics	Top 20 topics	Top 25 topics
Amazon	0.956	1.600	2.077	2.403	2.476
ProQuest	1.056	1.773	2.231	2.381	2.482
Reddit	1.504	2.129	2.383	2.633	2.701
YouTube	1.490	2.174	2.524	2.739	2.830

Table 5
Results of pairwise comparisons with the sentiment volume by topic as the independent variable.

Compare	Chi-square statistic	Degrees of freedom	Adjusted alpha	Original P-value
Amazon-ProQuest	27.918	22	0.008	0.000(***)
Amazon-Reddit	32.251	22	0.008	0.000(***)
Amazon-YouTube	33.006	22	0.008	0.000(***)
ProQuest-Reddit	1.651	22	0.008	1.000
ProQuest-YouTube	4.589	22	0.008	0.193
Reddit-YouTube	1.182	22	0.008	1.000

*** Significant difference observed.

products or services and analyze their competitive position.

Second, ProQuest (a platform for online mass media) is biased toward a few topics (e.g., topic 3 [OS Security Updates], topic 7 [Tablet Market Trends and Sales], and topic 13 [Seasonal Shopping and Deals] in the case study), with rich information limited to positive sentiments. Online mass media frequently promotes brands by engaging target consumers. Therefore, ProQuest effectively analyzes the positive aspects of product and service offerings, particularly for comparative analysis, although the topics to be covered can be limited. Unlike other platforms, Reddit (a platform for collaborative online social media) has negative sentiment information with rich trends and semantic information. It is a platform for effectively sharing and communicating information with others for various purposes, which makes it rich in information.

Finally, YouTube, a platform for creative output and online social media, was found to have the highest subject diversity, which is generally viewed as a positive aspect. Like ProQuest, YouTube is the preferred platform for opinion leaders who want to experience new products and services in advance and share their experiences. Although positive and negative experiences can be communicated, positive experiences dominate the negative experience, given the opinion leaders' expectations of new products and services. Indeed, they have been used to promote products and services. However, YouTube shows relatively low time series consistency, indicating that events can significantly influence mentions of topics.

Regarding the semantic diversity factor, the subjects actively discussed differed by media. For example, the topic of great interest in one media platform was the least focused on another. Thus, various media must be analyzed for firms to derive brand intelligence from multiple perspectives. Regarding the sentiment factor, two subfactors were considered: sentiment volume and sentiment polarity. Sentiment volume was high only in Amazon (e-commerce-based media). Sentiment polarity was positive in all media except Reddit (collaborative social media). Thus, it should be noted that relying on a single media platform to analyze UGC may offer biased analysis results. As for the trend consistency dimension, Amazon (*E*-commerce-based media) and Reddit (collaborative social media) had more information than the others, which is useful for predicting the changes in brand-related issues.

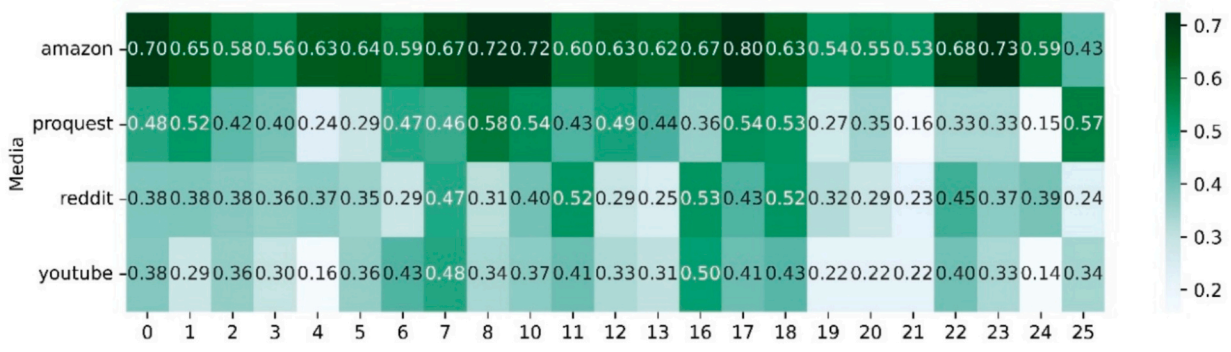


Fig. 6. Sentiment volume factor information completeness heatmap by brand-related media.

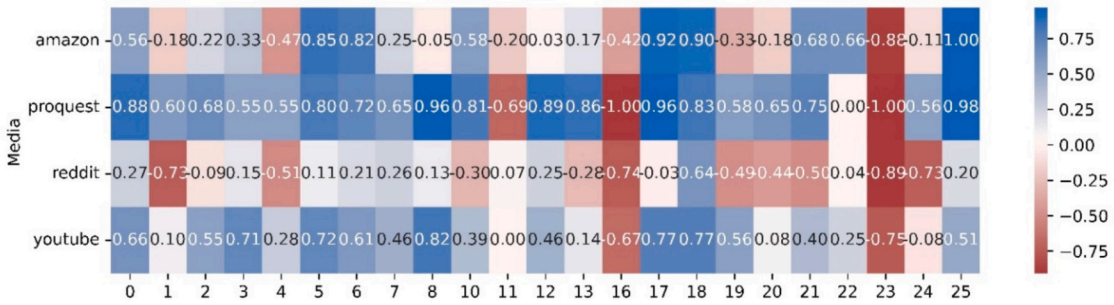


Fig. 7. Sentiment polarity factor information completeness heatmap by brand-related media.

Table 6
Results of pairwise comparisons with the sentiment polarity by topic as the independent variable.

Compare	Chi-square statistic	Degrees of freedom	Adjusted alpha	Original *P*-value
Amazon-ProQuest	3.528	22	0.008	0.361
Amazon-Reddit	5.270	22	0.008	0.130
Amazon-YouTube	0.478	22	0.008	1.000
ProQuest-Reddit	16.252	22	0.008	0.000(***)
ProQuest-YouTube	6.163	22	0.008	0.078
Reddit-YouTube	12.824	22	0.008	0.002(**)

** $p < 0.05$ (significant at the 5% level).
** $p < 0.01$ (significant at the 1% level).
*** $p < 0.001$ (significant at the 0.1% level).

5.2. 5.2. Theoretical and managerial implications

The study provides a theoretical foundation for either combining multiple media platforms or selecting the appropriate one based on the objective, from the perspective of information completeness. Although previous studies have primarily focused on single analytical methods, this research broadens the understanding of the informational characteristics of media data by comparing these traits across various media within the theoretical framework of information completeness. Specifically, it compares and analyzes the informational characteristics of four types of media—e-commerce-based media, online mass media, collaborative social media, and creative-outlet social media—through the lens of completeness, thereby theoretically supporting the importance of media type in UGC analysis.

From a methodological perspective, this study is one of the earliest attempts to compare the informational characteristics of three or more types of brand-related media across multiple dimensions, and it is expected to serve as a foundation for subsequent research on deriving brand intelligence from media. Through the proposed methodology, it becomes possible to quantitatively compare and analyze the differences in semantic diversity, sentiment, and temporal consistency information that may arise when each platform has the same amount of information.

The research findings also have managerial implications. This study emphasizes that various brand media platforms have distinct informational characteristics and that relying on a single media platform can lead to biased results. For instance, if the potential for positive bias in online mass media and creative social media is not recognized, it could result in a misjudgment of the actual market position. This study highlights the importance for companies to strategically choose media platforms when analyzing UGC data. Because collecting and analyzing data from all platforms can be resource intensive, it is crucial for companies to understand the informational characteristics of each media platform and use them strategically according to the specific needs of the situation.

The information provided by the platform differs significantly, indicating the possibility of biased conclusions being drawn if we rely only on a single platform. The best approach is to combine various brand-related media to extract intelligence, but this may cost a lot of time and effort. Accordingly, the strategic use of brand-related UGC media is required, depending on the situation. Although we investigated only four media platforms empirically, we regarded them as representing the characteristics of four media platform types: e-commerce-based media, online mass media, collaboration social media, and creative-outlet social media.

First, we found that no single platform covers all topics addressed by the four platforms; the information that each of them has is so heterogeneous that relying only on a single platform can yield biased results. In particular, the information from e-commerce-based media is more limited than the others because this media platform is concerned mainly with early-stage UX. Nevertheless, user needs arise during the entire life cycle of products and services and thus need to be extracted from various media platforms to reflect diverse perspectives on innovation activities.

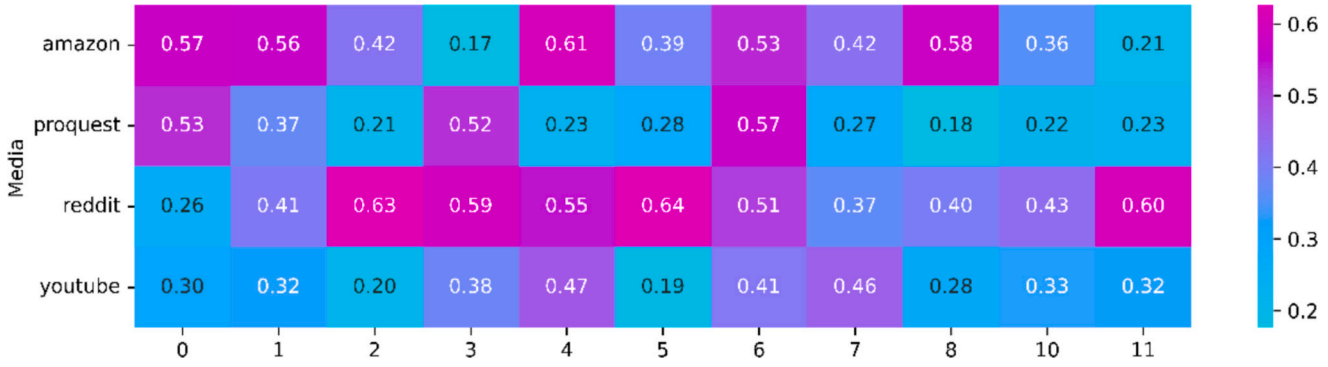


Fig. 8. Trend consistency factor information completeness heatmap by brand-related media.

Table 7
Summary of information characteristics by platform.

		Amazon (E-commerce-based media)	ProQuest (Online mass media)	Reddit (Collaboration Social media)	YouTube (Creative-outlet Social media)
Semantic diversity	**Rich** (Proportion ≥0.5)	#1; #22; #16 Battery and Charging; Customer Service and Returns Process; Gifts for Family Celebrations	#3; #7; #13 OS Security Updates; Tablet Market Trends and Sales: Seasonal Shopping and Deals	#5; #8; #11; #23 Digital Art and Notetaking Tools; Competitor characteristics; Feedback and Advice; Technical troubleshooting	–
	Overall diversity (Topic size = 25)	Narrow 2.476	Narrow 2.381	Diverse 2.701	Diverse (2.83)
Sentiment	**Volume** (Average)	Abundant (0.628)	Moderate (0.406)	Moderate (0.372)	Moderate (0.335)
	Polarity (Average)	Generally positive (0.224)	Mostly positive (0.502)	Generally negative (−0.14)	Generally positive (0.337)
Trend consistency	**Density** (Average)	Abundant (0.438)	Rare (0.328)	Abundant (0.489)	Rare (0.333)

Note: # Numbers in the table indicate topic numbers.

Second, e-commerce-based media is rich in sentiment information, thus helping monitor consumers' satisfaction through opinion mining. However, it should be noted that the topics addressed within this media may be limited and lack semantic and trend information. In particular, newly emerged issues may be missed because of the lack of trend information, and changes in user needs may not be adequately captured. Therefore, this medium is suitable for examining early signals or warnings in the market and positive or negative evaluations of products and services.

Third, online mass media and creative-outlet social media generally have positive sentiments and thus are valuable for understanding the advantages of a brand. A better marketing strategy can be established once a brand's strengths are well understood. However, relying too much on these media types can cause a firm to overestimate its brand because positive sentiment dominates them. In addition, seeing that the distribution of semantic information over the topics is similar, these two media platforms are likely to be alternatives rather than complementary.

Finally, adopting collaborative social media to facilitate innovation, such as creating diverse ideas for new product development, is worthwhile. This media platform seems to cover many topics, providing rich semantic information and being characterized as negative sentiments. Thus, it is beneficial for effectively extracting customer needs, specifically customer complaints. With sufficient trend information, this platform can be used to monitor customer needs over time.

6. Conclusions

Advances in NLP technologies have led to extensive research on identifying customer needs and requirements from online media UGC. Simultaneously, the explosive growth of online media has increased its diversity and complexity. However, most of these studies focused on the process of deriving intelligence from a single medium; relatively few attempts have been made to compare the differences among the media, which affects the nature of the intelligence extracted from the media. This study proposes a quantitative approach to comparing the informational characteristics of media and reveals that brand-related media have their own distinguishing features in the information they provide, particularly, in terms of semantic diversity, sentiment, and trend consistency.

Despite its valuable contributions, this study has several limitations. First, our analysis focused exclusively on assessing the completeness of information within a single media by examining semantic diversity, sentiment, and trend consistency. It is important to recognize that various other methodologies and perspectives exist for evaluating the informational characteristics of UGC. For instance, beyond the three dimensions mentioned, the completeness of information can also be evaluated in terms of the reliability of the information and its geographical scope. However, a comparison was not feasible due to the lack of publicly available information across all media in this study. Similarly, the quality of information encompasses various dimensions, including the accuracy and timeliness of the information. Further studies can be conducted to investigate the informational characteristics of UGC from other perspectives.

Second, we examined UGC diversity across platforms with different purposes and substantial brand-related content. Nevertheless, only four UGC platforms concerning one product were examined in this study. Future research should explore a wider range of platforms and products to enhance the generalizability of these findings.

Third, this study addressed information imbalance across platforms by applying sampling; however, it assumed an equal volume of information across media, which may not reflect real-world conditions. However, differences in information volume could also be seen as a characteristic of each platform, worth being examined in future studies.

Fourth, a deeper exploration of the causes behind the informational differences across platforms is necessary. Potential reasons for these differences could include the primary user base of each platform, the platform's UI structure or system, and the presence of lead users. Understanding these factors could reduce bias in brand UGC analysis and provide broader insights.

Finally, from a practical perspective, future research should focus on developing methodologies that enable companies to strategically select brand UGC platforms based on specific contexts. This requires preliminary investigations into how companies analyze UGC platforms, which can be conducted through interviews or literature reviews.

CRediT authorship contribution statement

Sanghyun Park: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Sungjoo Lee:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Acknowledgement

This work was supported by National Research Foundation of Korea (NRF) grant funded by the Korean Government (Ministry of Science and ICT) (RS-2022-NR070419). This work was supported by the SNU-Global Excellence Research Center establishment project.

Appendix A. Appendix

Table A1
Results of sentiment analysis applied to summaries by media.

Media type	Number of neutral document	Number of positive document	Number of negative document	Total
Amazon	3568(33.6 %)	4597(43.3 %)	2440(23.0 %)	10,605
ProQuest	6032(56.9 %)	4024(37.9 %)	549(5.2 %)	10,605
Reddit	6717(63.3 %)	1721(16.2 %)	2167(20.4 %)	10,605
YouTube	6951(65.5 %)	2735(25.8 %)	919(8.7 %)	10,605

Data availability

Data will be made available on request.

References

Amirmokhtar Radi, S., Shokouhyar, S., 2021. Toward consumer perception of cellphones sustainability: a social media analytics. Sustain. Product. Consump. 25, 217–233. https://doi.org/10.1016/j.spc.2020.08.012.

Bazzaz Abkenar, S., Haghi Kashani, M., Mahdipour, E., Jameii, S.M., 2021. Big data analytics meets social media: a systematic review of techniques, open issues, and future directions. Telematics Inform. 57 (October 2020), 101517. https://doi.org/10.1016/j.tele.2020.101517.

Bhimani, H., Mention, A.L., Barlatier, P.J., 2019. Social media and innovation: a systematic literature review and future research directions. Technol. Forecast. Soc. Chang. 144 (October 2018), 251–269. https://doi.org/10.1016/j.techfore.2018.10.007.

Brooks, S., 2015. Does personal social media usage affect efficiency and well-being? Comput. Hum. Behav. 46, 26–37. https://doi.org/10.1016/j.chb.2014.12.053.

Chang, Y.C., Ku, C.H., Chen, C.H., 2019. Social media analytics: extracting and visualizing Hilton hotel ratings and reviews from TripAdvisor. Int. J. Inf. Manag. 48 (April 2017), 263–279. https://doi.org/10.1016/j.ijinfomgt.2017.11.001.

Chaturvedi, I., Cambria, E., Welsch, R.E., Herrera, F., 2018. Distinguishing between facts and opinions for sentiment analysis: survey and challenges. Inform. Fusion 44, 65–77. https://doi.org/10.1016/j.inffus.2017.12.006.

Choi, J., Yoon, J., Chung, J., Coh, B.Y., Lee, J.M., 2020. Social media analytics and business intelligence research: a systematic review. Inf. Process. Manag. 57 (6). https://doi.org/10.1016/j.ipm.2020.102279.

Daugherty, T., Eastin, M.S., Bright, L., 2008. Exploring consumer motivations for creating user-generated content. J. Interact. Advert. 8 (2), 16–25. https://doi.org/10.1080/15252019.2008.10722139.

Goldberg, D.M., Abrahams, A.S., 2022. Sourcing product innovation intelligence from online reviews. Decision Supp. Syst. April 2022, 113751. https://doi.org/10.1016/j.dss.2022.113751.

Grootendorst, M., 2022. BERTopic: neural topic modeling with a class-based TF-IDF procedure. http://arxiv.org/abs/2203.05794.

Grosser, K.M., Hase, V., Wintterlin, F., 2019. Trustworthy or shady?: exploring the influence of verifying and visualizing user-generated content (UGC) on online journalism's trustworthiness. Journal. Stud. 20 (4), 500–522. https://doi.org/10.1080/1461670X.2017.1392255.

Hamed, K.H., Rao, A.R., 1998. A modified Mann-Kendall trend test for autocorrelated data. J. Hydrol. 204 (1–4), 182–196.

Haque, T. U., Saber, N. N., & Shah, F. M. (2018). Sentiment analysis on large scale Amazon product reviews. 2018 IEEE International Conference on Innovative Research and Development (ICIRD), 1–6.

He, W., Chen, Y., Tian, X., Chong, D., 2016. Actionable social media competitive analytics for understanding customer experiences. J. Comput. Inf. Syst. 56 (2), 145–155. https://doi.org/10.1080/08874417.2016.1117377.

Holsapple, C.W., Hsiao, S.H., Pakath, R., 2018. Business social media analytics: characterization and conceptual framework. Decis. Support. Syst. 110 (August 2016), 32–45. https://doi.org/10.1016/j.dss.2018.03.004.

James-Gilboe, L., 2005. The challenge of digitization: libraries are finding that newspaper projects are not for the faint of heart. Ser. Libr. 49 (1–2), 155–163. https://doi.org/10.1300/J123v49n01_06.

Jang, B., Yoon, J., 2018. Characteristics analysis of data from news and social network services. IEEE Access 6, 18061–18073.

Jelodar, H., Wang, Y., Rabbani, M., Ahmadi, S.B.B., Boukela, L., Zhao, R., Larik, R.S.A., 2021. A NLP framework based on meaningful latent-topic detection and sentiment analysis via fuzzy lattice reasoning on youtube comments. Multimed. Tools Appl. 80 (3), 4155–4181. https://doi.org/10.1007/s11042-020-09755-z.

Jeong, B., Yoon, J., Lee, J.M., 2019. Social media mining for product planning: a product opportunity mining approach based on topic modeling and sentiment analysis. Int. J. Inf. Manag. 48, 280–290. https://doi.org/10.1016/j.ijinfomgt.2017.09.009.

Kaplan, A.M., Haenlein, M., 2010. Users of the world, unite! The challenges and opportunities of Social Media. Business Horizons 53 (1), 59–68.

Kietzmann, J.H., Hermkens, K., McCarthy, I.P., Silvestre, B.S., 2011. Social media? Get serious! Understanding the functional building blocks of social media. Business horizons 54 (3), 241–251.

Kim, J., Lim, C., 2021. Customer complaints monitoring with customer review data analytics: an integrated method of sentiment and statistical process control analyses. Adv. Eng. Inform. 49 (May), 101304. https://doi.org/10.1016/j.aei.2021.101304.

Ko, N., Jeong, B., Choi, S., Yoon, J., 2017. Identifying product opportunities using social media mining: application of topic modeling and chance discovery theory. IEEE Access 6, 1680–1693. https://doi.org/10.1109/ACCESS.2017.2780046.

Lee, I., 2018. Social media analytics for enterprises: typology, methods, and processes. Bus. Horiz. 61 (2), 199–210. https://doi.org/10.1016/j.bushor.2017.11.002.

Lee, S., Park, S., Kwak, M., 2022. Revealing the dual importance and Kano type of attributes through customer review analytics. Adv. Eng. Inform. 51 (August 2021), 101533. https://doi.org/10.1016/j.aei.2022.101533.

Li, Z., Zhang, X., Zhang, Y., Long, D., Xie, P., Zhang, M., 2023. Towards general text embeddings with multi-stage contrastive learning. ArXiv Preprint (ArXiv: 2308.03281).

Liu, Y., Jiang, C., Zhao, H., 2019. Assessing product competitive advantages from the perspective of customers by mining user-generated content on social media. Decis. Support. Syst. 123 (May), 113079. https://doi.org/10.1016/j.dss.2019.113079.

Loureiro, D., Barbieri, F., Neves, L., Espinosa Anke, L., & Camacho-collados, J. (2022). {T}ime{LM}s: diachronic language models from {T}witter. *Proceedings of the 60th annual meeting of the Association for Computational Linguistics: system demonstrations*, 251–260. Doi:10.18653/v1/2022.acl-demo.25.

Mariani, M.M., Nambisan, S., 2021. Innovation analytics and digital innovation experimentation: the rise of research-driven online review platforms. Technol. Forecast. Soc. Chang. 172 (January), 121009. https://doi.org/10.1016/j.techfore.2021.121009.

Mostafa, M.M., 2013. More than words: social networks' text mining for consumer brand sentiments. Expert Syst. Appl. 40 (10), 4241–4251. https://doi.org/10.1016/j.eswa.2013.01.019.

Nambisan, S., 2002. Designing virtual customer environments for new product development: toward a theory. Acad. Manag. Rev. 27 (3), 392–413. https://doi.org/10.5465/AMR.2002.7389914.

Ogunseye, S., Parsons, J., 2018. Designing for information quality in the era of repurposable crowdsourced user-generated content. *Advanced information systems engineering workshops: CAiSE 2018 international workshops, Tallinn, Estonia, June 11-15, 2018*. Proceedings 30, 180–185.

Parsons, J., & Wand, Y. (2014). A foundation for open information environments. *ECIS 2014 proceedings - 22nd European conference on information systems*, 0–9.

Pournarakis, D.E., Sotiropoulos, D.N., Giaglis, G.M., 2017. A computational model for mining consumer perceptions in social media. Decis. Support. Syst. 93, 98–110. https://doi.org/10.1016/j.dss.2016.09.018.

Reyes-Menendez, A., Saura, J.R., Filipe, F., 2020. Marketing challenges in the #MeToo era: gaining business insights using an exploratory sentiment analysis. Heliyon 6 (3). https://doi.org/10.1016/j.heliyon.2020.e03626.

Roma, P., Aloini, D., 2019. How does brand-related user-generated content differ across social media? Evidence reloaded. J. Bus. Res. 96 (November 2018), 322–339. https://doi.org/10.1016/j.jbusres.2018.11.055.

Sadvilkar, N., Neumann, M., 2020. {P}y{SBD}: pragmatic sentence boundary disambiguation. *Proceedings of second workshop for NLP open source software (NLP-OSS)*, 110–114. https://www.aclweb.org/anthology/2020.nlposs-1.15.

Shannon, C.E., 1948. A mathematical theory of communication. Bell Syst. Tech. J. 27 (3), 379–423. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x.

Sivarajah, U., Irani, Z., Gupta, S., Mahroof, K., 2020. Role of big data and social media analytics for business to business sustainability: a participatory web context. Ind. Mark. Manag. 86 (April 2019), 163–179. https://doi.org/10.1016/j.indmarman.2019.04.005.

StatCounter. Tablet browser market share worldwide. GlobalStats. https://gs.statcounter.com/browser-market-share/tablet/worldwide/.

Stieglitz, S., Dang-Xuan, L., 2013. Social media and political communication: a social media analytics framework. Social network analysis and mining 3, 1277–1291.

Stieglitz, S., Mirbabaie, M., Ross, B., Neuberger, C., 2018. Social media analytics – challenges in topic discovery, data collection, and data preparation. Int. J. Inf. Manag. 39 (October 2017), 156–168. https://doi.org/10.1016/j.ijinfomgt.2017.12.002.

Tsirakis, N., Poulopoulos, V., Tsantilas, P., Varlamis, I., 2017. Large scale opinion mining for social, news and blog data. J. Syst. Softw. 127, 237–248. https://doi.org/10.1016/j.jss.2016.06.012.

Wang, R.Y., Strong, D.M., 1996. Beyond accuracy: What data quality means to data consumers. J. Manag. Inf. Syst. 12 (4), 5–33.

Wang, W., Feng, Y., Dai, W., 2018. Topic analysis of online reviews for two competitive products using latent Dirichlet allocation. Electron. Commer. Res. Appl. 29 (January), 142–156. https://doi.org/10.1016/j.elerap.2018.04.003.

Xu, X., Wang, X., Li, Y., Haghighi, M., 2017. Business intelligence in online customer textual reviews: understanding consumer perceptions and influential factors. Int. J. Inf. Manag. 37 (6), 673–683. https://doi.org/10.1016/j.ijinfomgt.2017.06.004.

Zeng, D., Chen, H., Lusch, R., Li, S.H., 2010. Social media analytics and intelligence. IEEE Intell. Syst. 25 (6), 13–16. https://doi.org/10.1109/MIS.2010.151.

Zhan, Y., Han, R., Tse, M., Ali, M.H., Hu, J., 2021. A social media analytic framework for improving operations and service management: a study of the retail pharmacy industry. Technol. Forecast. Soc. Chang. 163. https://doi.org/10.1016/j.techfore.2020.120504.

Zhang, C., Tong, T., Bu, Y., 2019. Examining differences among book reviews from various online platforms. Online Inf. Rev. 43 (7), 1169–1187.

Zhu, Y.Q., Chen, H.G., 2015. Social media and human need satisfaction: implications for social media marketing. Bus. Horiz. 58 (3), 335–345. https://doi.org/10.1016/j.bushor.2015.01.006.

Sanghyun Park received his Master's degree on Industrial Engineering in the Department of Artificial Intelligence, Ajou University, Republic of Korea, in 2021. He is now a Ph.D candidate at the Department of Industrial Engineering, Seoul National University (SNU), Seoul, Republic of Korea. His research interests include new technology opportunities, patent analysis, and user-driven innovation.

Sungjoo Lee received Ph.D. on Industrial Engineering from Seoul National University in 2007. She is currently an Associate Professor of Industrial Engineering at SNU, Seoul, Republic of Korea. Before joining SNU, she worked as a faculty member of Ajou University, Seoul, Republic of Korea, and a visiting scholar at the Centre for Technology Management, Institute for Manufacturing, University of Cambridge, UK. Her research interests include technology roadmapping, patent engineering, and data-driven innovation.